

Fractions — Diagnostic Assessment

Ghanaian upper primary (B.S. 4-6), generated against a 10-node skill graph.

Name: _____ Class: _____
School: _____ Date: _____

Answer every question. For each item, **circle the letter** of the best answer. Working may be shown on the back of the page.

1. Compute $\frac{3}{4} \div \frac{1}{2}$.

- | | |
|------------------|------------------|
| A. $\frac{1}{2}$ | B. $\frac{3}{8}$ |
| C. $\frac{6}{4}$ | D. $\frac{8}{3}$ |

2. Convert $\frac{11}{4}$ to a mixed number.

- | | |
|--------------------|--------------------|
| A. $2 \frac{1}{4}$ | B. $2 \frac{3}{4}$ |
| C. $3 \frac{1}{4}$ | D. $\frac{11}{4}$ |

3. Simplify $\frac{12}{18}$.

- | | |
|------------------|------------------|
| A. $\frac{1}{2}$ | B. $\frac{2}{3}$ |
| C. $\frac{4}{6}$ | D. $\frac{6}{9}$ |

4. Compute $2 \frac{1}{2} \div 1 \frac{1}{4} \times \frac{2}{3}$.

- | | |
|------------------|--------------------|
| A. $\frac{1}{3}$ | B. 2 |
| C. $\frac{4}{3}$ | D. $\frac{25}{12}$ |

5. Compute $(\frac{2}{3} \times \frac{3}{4}) \div \frac{1}{2}$.

- | | |
|------------------|------------------|
| A. 1 | B. $\frac{1}{4}$ |
| C. $\frac{1}{2}$ | D. $\frac{1}{8}$ |

6. Which fraction is equivalent to $\frac{2}{3}$?

- | | |
|------------------|------------------|
| A. $\frac{2}{6}$ | B. $\frac{3}{4}$ |
| C. $\frac{4}{6}$ | D. $\frac{5}{6}$ |

7. Two friends each cut a kelewele plate. Kofi cuts his into 4 equal pieces and shades 1. Ama cuts hers into 4 pieces of different sizes and shades 1. Whose shaded part is exactly $\frac{1}{4}$ of the whole?

- | | |
|-------------|--------------|
| A. Both | B. Kofi only |
| C. Ama only | D. Neither |

8. Convert $2\frac{1}{3}$ to an improper fraction.

A. $\frac{2}{3}$

B. $\frac{3}{3}$

C. $\frac{6}{3}$

D. $\frac{7}{3}$

9. Yaa has $\frac{3}{4}$ kg of rice. She uses $\frac{2}{5}$ of it to cook jollof. How much rice did she use, in kg?

A. $\frac{5}{9}$

B. $\frac{5}{20}$

C. $\frac{6}{20}$

D. $\frac{5}{4}$

10. Compute $\frac{4}{6} \times \frac{3}{8}$ and give your answer in lowest terms.

A. $\frac{1}{4}$

B. $\frac{1}{3}$

C. $\frac{7}{14}$

D. $\frac{12}{48}$

11. Ama bought $3\frac{3}{4}$ kg of yam at the market. Express this as an improper fraction in kg.

A. $\frac{7}{4}$

B. $\frac{12}{4}$

C. $\frac{13}{4}$

D. $\frac{15}{4}$

12. Compute $\frac{1}{2} \times \frac{2}{3} \div \frac{1}{4}$.

A. $\frac{1}{3}$

B. $\frac{1}{12}$

C. $\frac{3}{4}$

D. $\frac{4}{3}$

13. Kofi has 15 baskets of mangoes. Each basket holds 30 mangoes. How many mangoes does he have in total?

A. 45

B. 150

C. 450

D. 4500

14. Compute $\frac{2}{3} \times \frac{4}{5}$.

A. $\frac{6}{8}$

B. $\frac{8}{8}$

C. $\frac{8}{15}$

D. $\frac{10}{12}$

15. Compute $1\frac{1}{2} \times \frac{2}{3} \div \frac{1}{4}$.

A. $\frac{1}{4}$

B. 1

C. 4

D. $\frac{3}{8}$

16. Akua walked $\frac{17}{5}$ km from her home to the market. Write this distance as a mixed number.

A. $2\frac{2}{5}$

B. $3\frac{2}{5}$

C. $3\frac{3}{5}$

D. $5\frac{3}{5}$

17. Ama wants to share 17 oranges equally among 4 friends. How many oranges does each friend get, and how many are left over?

A. 3 each, 5 left over

B. 4 each, 0 left over

C. 4 each, 1 left over

D. 5 each, 3 left over

18. Kwame has $\frac{6}{8}$ of a tray of meat pies. He divides his portion into pieces, each one $\frac{1}{4}$ of a full tray. How many pieces does he get?

A. $\frac{1}{3}$

B. 3

C. $\frac{6}{32}$

D. $\frac{32}{6}$

19. Compute $1\frac{1}{2} \times 2\frac{1}{2}$ and write the answer as a mixed number.

A. $2\frac{1}{2}$

B. $3\frac{3}{4}$

C. $\frac{5}{4}$

D. $\frac{15}{4}$

20. A rectangle is divided into 5 equal parts. 2 of those parts are shaded. What fraction of the rectangle is shaded?

A. $\frac{2}{3}$

B. $\frac{2}{5}$

C. $\frac{3}{5}$

D. $\frac{5}{2}$

21. Esi wrote $\frac{12}{18}$ on her exercise book. Express this as a fraction with denominator 3.

A. $\frac{1}{3}$

B. $\frac{2}{3}$

C. $\frac{4}{3}$

D. $\frac{6}{3}$

22. Fill in the missing number: $\frac{3}{5} = \frac{?}{15}$

A. 8

B. 9

C. 10

D. 13

23. Simplify $\frac{6}{9}$ to its lowest terms.

A. $\frac{1}{3}$

B. $\frac{2}{3}$

C. $\frac{3}{4}$

D. $\frac{6}{9}$

24. What is 6×7 ?

A. 13

B. 40

C. 42

D. 48

25. What is $3 \div 6$?

A. 0

B. 0.5

C. 2

D. Cannot be done

26. Compute $1\frac{3}{4} \times \frac{2}{3}$.

A. $\frac{2}{3}$

B. $\frac{7}{6}$

C. $\frac{5}{7}$

D. $\frac{21}{12}$